B1 statute-wise GRU memory: Methodology and Evidence

CS781 Capstone – Krishna K Karthikeya G N

12 November 2026

Status. Post-run methodology reconstruction grounded in archived source and reports. This document is not an executed historical notebook.

Research question

Test whether recurrent memory can accumulate label-specific evidence across the windows of a long judgment. The recurrent head can model order across chunks, unlike a purely independent max or mean pool, but adds trainable complexity on top of a small per-label signal.

Architecture and mechanism

B1 encodes four windows; statute-wise gated recurrent units update a memory for each label, and a global residual head supplements the local evidence stream.

Measured result

Scored submission	1016621
Codabench macro-F1	0.2238
Archived local metric	0.2293 official-dev macro-F1

Local official-dev macro-F1 is 0.2293 and Codabench is 0.2238. More head capacity alone did not improve the observed result; the report cannot prove whether optimization or evidence placement was the dominant cause.

Limitations and next check

The recurrent path processes a bounded number of chunks; it is not full-document attention. Local, pilot, and hidden metrics are deliberately kept distinct. A single scored run does not establish a causal architecture effect or statistical significance.

Source and reproducibility

Original source modules: phase5_statute_gru.py, phase5_multiwindow.py, phase5_b1_train.py. SHA-256 hashes, the original report, and the prediction ZIP receipt are in the companion provenance file. The explanatory notebook was written after the historical run; original execution output is never implied. Reproduction requires authorized course data, the Phase 1 parent weights/tokenizer, and the dependencies in the archived source.

Documentation disclosure

AI assistance was used to organize the archived evidence and draft this explanatory document. All numerical claims above come from saved run records or the provided finished-submission screenshot; missing evidence is labelled as such.